Outputs Explanation

Which genes track the shape of a neuron? Gene expression and morphology in inhibitory neurons of mouse motor cortex (Scala et al. 2021 Patch-seq data). This document explains every file written to the outputs folder by analysis.py: what each one is for, how to read it, what it represents, what can be concluded from it and why, and how the result compares with what the original authors reported.

Problem Statement

The analysis uses 358 inhibitory neurons. For each neuron three things are known: its family (Pvalb, Sst, Vip or Lamp5, a label the original authors assigned from its genes), its shape (summarised as five numbers: dendrite size and reach, axon size and spread, dendrite branching, axon branching, axon vertical extent) and its gene activity (1,500 numbers, one per gene). The question is whether the gene numbers tell us the shape numbers. The complication, which runs through every output, is that the family label already tells us a great deal about both the genes and the shape, so the analysis always asks what genes add once family and position in the cortex are already known.

1. results_gene_associations.csv

Purpose

This is the gene list the brief asked for, in full. It records the result of 7,500 separate attempts, one for every pairing of a gene (1,500) with a shape number (5). Each attempt asks: among cells of the same family, at the same depth and with the same sequencing depth, do cells with more of this gene have more, or less, of this shape?

How to read it

One row per attempt, sorted with the strongest at the top. The important columns are: gene and axis (which gene, tested against which shape); adjusted_effect (how far the shape moves, in standard deviations, per standard deviation of the gene, after family and depth are accounted for; negative means the gene is lower in cells with more of that shape); q_value (the evidence corrected for having run 7,500 tests; the number to trust; below 0.05 would count as a finding); bootstrap_stability (out of 100 resamplings of the cells, the share in which a sparse model kept this gene; 0.70 or above would count as robust); selected_by_baseline (whether the authors' approach, which ignores family, also picked this gene); known_family_marker (whether it is one of the well-known family defining genes).

Example, the first row: Tmem255a against axon size, effect of minus 0.53, q of 0.09, stability of 0.59. Cells with more Tmem255a have smaller axons, the result would be a fluke about 9 times in 100, and the gene was kept in 59 of 100 resamplings. Close, but below the bar on both counts.

What it represents and what it tells us

Nothing in the table reaches q below 0.05, and no gene reaches 0.70 stability. The best candidates are Tmem255a, Penk, Igfbp4 and Lama4 for axon size, Col16a1 and Qpct for axon vertical extent, and Clu for dendrite size. Two kinds of gene appear among them: genes that plausibly build shape (Lama4 is a laminin that growing axons crawl along; Fmn1 builds the actin filaments that push out branches; Ackr3 is a guidance receptor) and genes that mark subtypes within a family (Penk, Kit, Cnr1). The second kind is a warning that some finer identity still hides inside the four families.

Conclusion

Within families, no gene is convincingly linked to any aspect of shape. The conclusion follows from the q values: with 7,500 tests, a few will always look impressive by luck, and the correction shows the best results are consistent with luck. The leads are the first genes to test in a larger dataset, not findings.

Comparison with the authors

The authors did not test genes one by one against morphology, so there is no direct comparison. Their regression from genes to electrical behaviour selected well known family marker genes, and the fact that our baseline (step 4) picked Pvalb, Sst, Vip and Lamp5 among its genes matches that.

2. results_model_summary.csv

Purpose

The scoreboard: one row per shape number, with every statistic the analysis computed. This file tells us how sure the result is.

How to read it, column group by column group

The three prediction accuracy columns (r2_covariates_only, r2_genes_only_baseline, r2_covariates_then_genes) each give an R squared, a number from 0 to 1, where 0 means no better than guessing the average cell and 1 means perfect. The three are for three predictors tested on cells they had not seen: one told only the family and depth; one told only the 1,500 genes (the authors' approach); one told both. Read across a row. For axon size the three are 0.40, 0.41 and 0.40. They are the same on every row.

agreement_genes_vs_covariate_predictions: for every cell, what the genes only predictor guessed is compared with what the family and depth predictor guessed; the column is the correlation between the two lists of guesses, 0.75 to 0.89. The two predictors make nearly the same guesses.

gene_r2_on_residual, permutation_null_95th_percentile, permutation_p_value: after removing what family and depth explain, how much of the leftover shape genes explain (essentially zero), what scrambled data typically gives, and how often scrambled data did as well as the real data (for dendrite size, 83 percent of the time).

genes_at_q_below_threshold (0 on every row), pi0_share_of_null_tests (0.91, the estimated share of the 7,500 tests where the gene truly has no relationship to the shape), robust_genes (0).

effect_correlation_confidence_0.6, 0.7, 0.8 and the matching cell counts: the analysis repeated on only the well labelled cells (274, 237 and 189 cells); the effect sizes agree with the full run at 0.98, 0.97 and 0.95, so the result is not caused by mislabelled cells.

continuity_mantel_r and continuity_p for each family: within one family, whether pairs of cells that are similar in genes are also similar in shape. Sst 0.10 (p 0.01) and Lamp5 0.17 (p 0.005) are above chance; Pvalb and Vip are not. These values repeat on every row because they are per family, not per shape.

What it represents and what it tells us

A plain analogy: predicting a person's shoe size. A predictor told only adult or child scores some amount. A predictor told the person's entire medical record scores the same amount. Told both, still the same. The medical record was useful only because it revealed adult or child. Here, the genes are useful only because they reveal the family.

Conclusion

Genes predict shape exactly as well as the family label does and not a bit more; the gene model is, in effect, a family detector. No individual gene survives correction, and this is not because of poor labels. In two families there is a faint whole genome echo of shape that no single gene carries. The conclusion follows because every one of those statements has a number and a null next to it in this table.

Comparison with the authors

The authors reported that their sparse regression from genes to morphology supported the separation of major families. This table says the same, and adds that the regression supports only the separation of families. The authors' accuracy for genes to electrical behaviour was 0.38; ours for genes to shape is 0.26 to 0.41, the same order of magnitude. The authors argued that variation within families is continuous; the Mantel results find such continuity for shape in Sst and Lamp5, the two families the paper itself describes as internally graded, but not in Pvalb, where the paper described a smooth change in one specific axon feature within layer 5. Our test uses all five shape numbers across all layers after removing depth, so it is a broader and stricter test than theirs; a targeted repeat on layer 5 Pvalb cells and that one feature would be the direct comparison.

3. results_function.csv

Purpose

The function deliverable. Genes are switched on and off by other genes called transcription factors. A database (CollecTRI) lists, for 668 of them, which genes each one controls. From that, an activity score was computed for each factor in each cell, and the same family, same depth test was run against each shape number. The file also holds any gene set enrichment results.

How to read it

Five rows, the five strongest factors, with the same columns as the gene table. Smad4, Smad3 and Jun are all lower in cells with larger dendrites, q of 0.15. The kind column says these are the strongest five, not significant findings. There are no enrichment rows.

What it represents and what it tells us

No regulator is linked to shape within families at the usual bar. Smad4 and Smad3 (TGF beta signalling) and Jun (AP 1 signalling) are known to be involved in dendrite growth, so the direction is believable, but the evidence is below the bar.

Conclusion and why the enrichment is empty

Enrichment analysis asks what the genes on a list have in common; it needs a list of genes that passed. The list is empty, so the analysis has nothing to run on. Running it anyway on genes that failed would produce an impressive looking table that means nothing. An empty result is the honest answer.

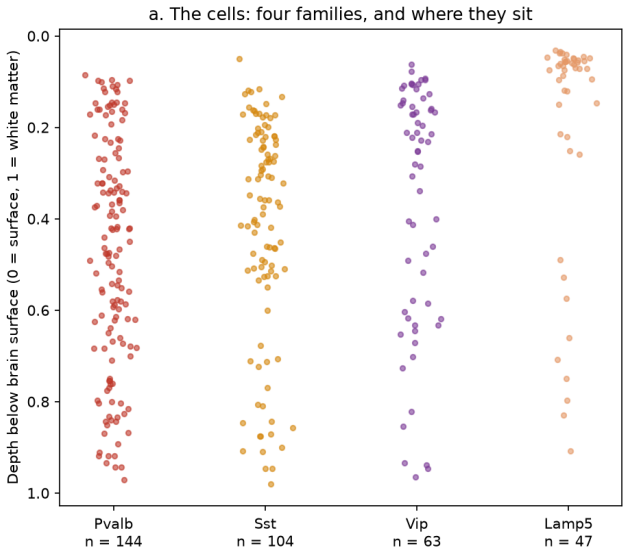
Comparison with the authors

The authors did not analyse transcription factor activity. Their main regression selected ion channel genes alongside markers, which they noted as mechanistically interesting; the equivalent lead here is the Smad and Jun signalling factors with dendrite size, equally unproven.

4. figure_main.png

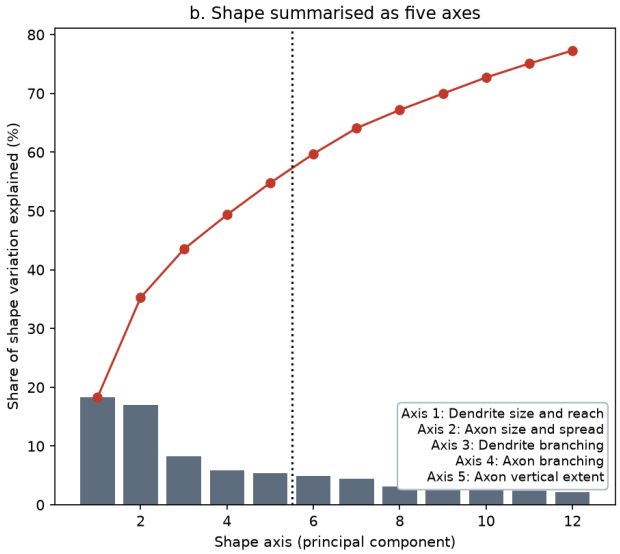
The six panels tell the whole argument in one page. Read them left to right, top row then bottom row.

Panel a. The cells: four families, and where they sit



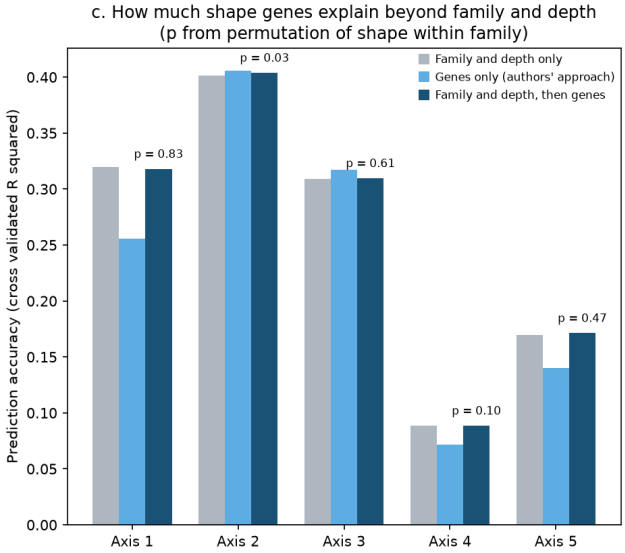
One dot per neuron, grouped by family, placed at its depth (brain surface at the top, white matter at the bottom). The counts are 144 Pvalb, 104 Sst, 63 Vip and 47 Lamp5. What to notice: nearly every Lamp5 cell sits at the very top. Family and depth go together, so if only one were corrected for, the other would sneak back in. This is why both are covariates in every model.

Panel b. Shape summarised as five axes



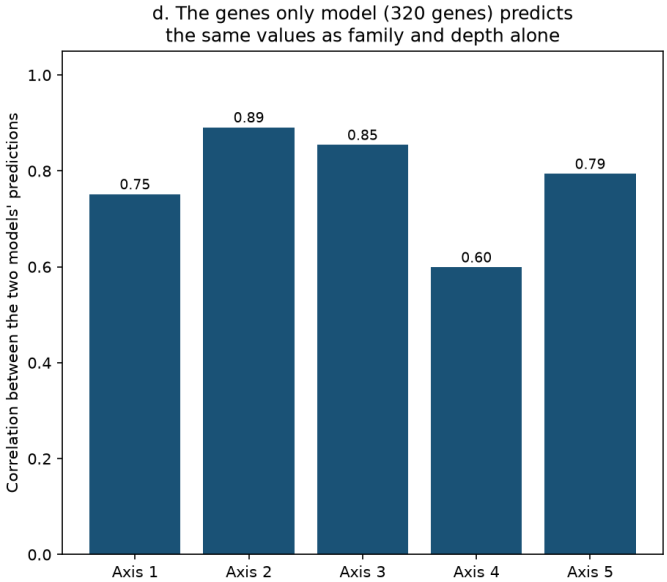
The 49 shape measurements were squeezed into directions of difference by principal component analysis, biggest first. The grey bars show how much of the total shape variation each direction captures: 18, 17, 8, 6 and 5 percent for the first five, then under 5 each. The red line is the running total, reaching 55 percent at five. The dotted line marks where we stopped, and the box names the five. The reason for five is visible: after the fifth bar the heights flatten out, so each further axis buys little and costs 1,500 more tests.

Panel c. How much shape genes explain beyond family and depth



The central panel. For each shape number, three bars: how well family and depth alone predict it (grey); how well genes alone predict it (light blue, the authors' approach); how well both together predict it (dark blue). On every shape the three bars are level. The p above each dark bar is from the scrambling test: after removing what family and depth explain, the little that genes still explain was compared with what scrambled data gives, 200 times; the p is how often scrambled data did as well. For dendrite size, 83 percent of the time. For axon size, 3 percent, but the amount explained there is three thousandths of the variation, detectable but not meaningful. Conclusion: genes predict shape no better than the family label.

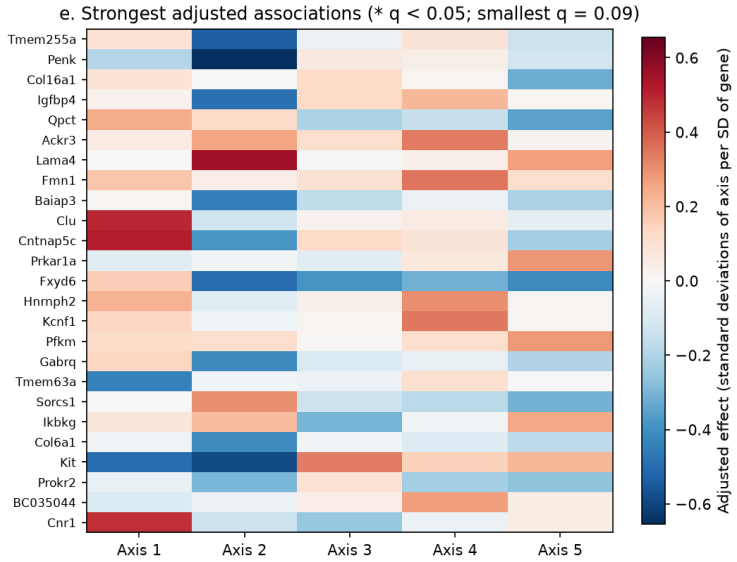
Panel d. The genes only model predicts the same values as family and depth



One bar per shape number: how closely the genes only model's cell by cell guesses match the family and depth model's guesses, as a correlation from 0 to 1. The values are 0.75, 0.89, 0.85, 0.60 and 0.79. The two models make nearly the same guesses, which means the gene model has learned to recognise the family and read off its typical shape. The lowest value, 0.60, is for axon branching, which no model predicts well (all bars under 0.1 in panel c), so both models are mostly guessing noise there. This panel answers the objection that the gene

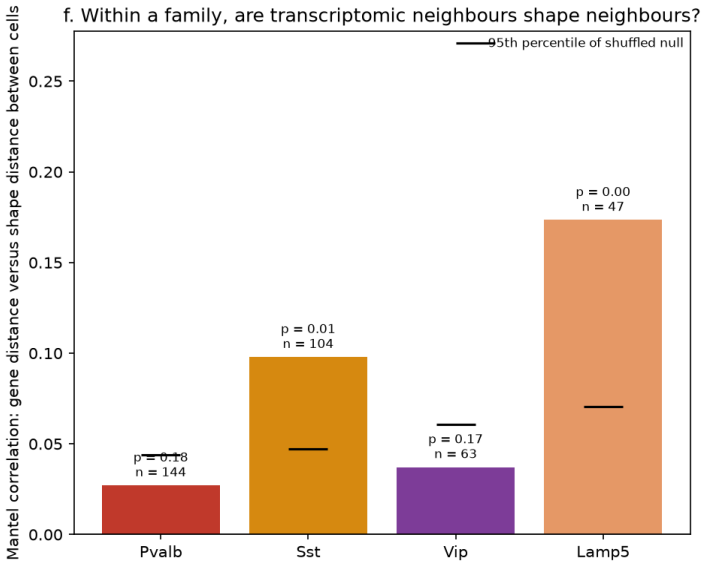
model picked only nine famous family markers out of 320: it does not need the famous ones, because hundreds of genes rise and fall with family.

Panel e. Strongest adjusted associations



A grid of the 25 genes with the strongest within family associations (rows) against the five shape numbers (columns). Red means the gene goes up with that shape, blue means down, deeper colour means a bigger effect. For example Lama4 is dark red on axon size (more Lama4, bigger axon) and Penk is dark blue on the same axis (more Penk, smaller axon). Stars would mark anything passing the corrected bar of q below 0.05; there are none, and the title states that the best q is 0.09. The panel shows that the search was done and names the leads.

Panel f. Within a family, are transcriptomic neighbours shape neighbours?



One bar per family: within that family, the correlation between how different two cells are in genes and how different they are in shape, across all pairs of cells. The black tick is the 95th percentile of what 200 scramblings produced; a bar above its tick is above chance. Sst (0.10, p 0.01) and Lamp5 (0.17, p 0.005) are above chance; Pvalb and Vip are not. These are small correlations, as pairwise distance correlations always are, but they are

real, and they are the paper's within family continuity idea tested for shape at the level of individual cells. They do not contradict panel e: a gradient carried by hundreds of genes at tiny effect sizes shows up in a distance test and still leaves no single gene significant.

5. analysis_log.txt

Everything printed during the run: how many cells survived each filter, what each shape axis is made of, how many genes were kept, every accuracy score, the ten strongest genes, the permutation progress and the five strongest transcription factors. It is the audit trail. Every number on the slides and in this document can be traced to a line in it.

Conclusion

In these data, gene expression predicts the shape of an inhibitory neuron by predicting its family. Inside a family, any gene level signal is too faint for 358 cells to resolve, although a faint whole genome correspondence survives in Sst and Lamp5.